



Society of Petroleum Engineers

SPE-227348-MS

Overcoming Conventional Limitations: An Advanced Emulsified Filtercake Removal Fluid for Extended Reaction Time

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227348-MS](https://doi.org/10.2118/227348-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

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Abstract

Water-based reservoir drill-in fluids (RDFs) often deposit a filtercake composed of polymers (such as starch or biopolymers) and acid-soluble bridging agents (e.g., calcium carbonate) on the wellbore. Effective removal of this filtercake prior to production is critical to avoid formation damage and impaired well performance. Conventional filtercake cleanup methods have significant limitations under high-temperature conditions. Strong acids (e.g., HCl or organic acids) react too rapidly and non-uniformly, creating localized breakthroughs and leaving portions of the cake intact. Enzyme-based breakers offer gentler, more uniform polymer degradation but historically lack thermal stability at high temperatures. Chelating agents can slowly dissolve CaCO_3 with less corrosion but their efficiency drops and reaction rate accelerates undesirably at elevated temperatures. Acid precursors that hydrolyze to release acid over time have shown promise for delayed reaction, yet conventional designs fail to maintain sufficient delay beyond ~24 hours at temperatures above 200 °F (93 °C). In practice, many high-temperature wells have struggled to achieve uniform filtercake removal because available breaker fluids either react too quickly or degrade under prolonged heat exposure.

This paper presents a novel emulsified fluid system for filtercake removal designed specifically to overcome high-temperature limitations. The innovation lies in extending the application of emulsified acids from matrix acidizing to wellbore cleanup and filtercake dissolution. By incorporating enzyme into the emulsified organic acid formulation, the system targets both major filtercake components: the enzyme slowly degrades polymeric binders, while the emulsified organic acid provides controlled, delayed dissolution of the CaCO_3 bridging solids.

Laboratory filtercakes were generated using lab-prepared water-based mud (WBM), comprising a low-solids, non-dispersed, water-soluble polymer-based drilling mud with acid-soluble bridging and weighting agents. Thermal stability tests were conducted on the new emulsified filtercake removal fluid at 200 °F for various periods of time by aging the fluid in sealed high-pressure/high-temperature (HP/HT) cells at 200 °F and examining any phase separation. Dissolution efficiency tests were then performed by exposing the lab-prepared filtercake samples to the emulsified filtercake removal fluid at 200 °F for 1, 3, and 5 days at a ratio of 1:10 filtercake solids to solution and aperiodic solubility measurements were taken to quantify the extent of filtercake dissolution at each interval. Corrosion testing was conducted by immersing C95 steel coupons in the fluid at 200 °F to measure coupon weight loss and corrosion rate.

Results demonstrate that the emulsified filtercake removal system achieved satisfactory dissolution efficiency and improved reaction longevity. The reaction was effectively delayed and controlled which can prevent premature breakthrough and fluid loss. The dissolution increased gradually over days and by the 5-days mark, the emulsified filtercake removal fluid achieved around 50% dissolution of the filtercake, in stark contrast to conventional filtercake removal treatments that would have either reacted instantaneously or ceased functioning by this time. The enzyme was able to operate at 200 °F, breaking down starch in the filtercake as evidenced by an iodine indicator test (no purple staining after treatment, indicating starch elimination). The emulsified acid remained stable throughout the soaking period, releasing acid gradually for sustained dissolution. Thermal stability tests showed that the fluid was stable with no significant phase separation or degradation after 5 days at 200 °F. Corrosion impact on C95 steel was minimal and the fluid exhibited a low corrosion rate, attributed to the acid being sequestered in the oil phase.

In summary, the developed emulsified filtercake removal system provides a single-step, high-temperature filtercake breaker that offers extended reaction time and uniform cleanup, overcoming the 24-hour limit of conventional systems.

Introduction

Open-hole completions drilled with reservoir drill-in fluids (RDFs) rely on an intentionally formed filtercake to limit fluid losses into the formation. This filtercake is a thin, low-permeability layer deposited on the wellbore wall, composed of drilling fluid solids such as weighting or bridging agents and polymers. In water-based mud (WBM) systems, the filtercake typically contains calcium carbonate (acid-soluble bridging particles) and polymeric filtrate reducers like starch or xanthan gum. While the filtercake is beneficial during drilling, it becomes undesirable once the well is completed because it can impede hydrocarbon flow and cause high skin if left intact. Therefore, effective removal of the filtercake before production or injection is critical to maximize well performance ([Gianna, et al. 2022](#)).

An ideal filtercake removal fluid should slowly degrade or dissolve the filtercake in a controlled manner across the entire interval. If the treatment reacts too quickly or aggressively, it can create a localized "breakthrough" path the point of initial contact, causing the filtercake removal fluid to leak off into the formation and bypass the rest of the filtercake. This leads to incomplete removal, with patches of filtercake remaining. Such non-uniform cleanup can result in isolated production zones, or premature water breakthrough ([Gianna, et al. 2022](#)). Conversely, if the treatment is too inert or slow, it might not dissolve the filtercake within the available well completion time. The treatment design thus requires a balanced approach and a sufficient delay to place the treatment, specifically in multi-lateral clean-up operations to avoid the fluid losses before the end of the operation.

Conventional Filtercake Removal Methods and Limitations

Over the years, a variety of chemicals have been developed to address different components of WBM filter cakes. Key categories of conventional filtercake removal systems will be discussed. Each of these conventional approaches addresses part of the components of the filtercake and has its own reaction duration limitation.

Strong Acids

Mineral acids like HCl or organic acids (formic, acetic) can dissolve calcium carbonate bridging material very effectively and also hydrolyze some polymers. Typically, a 5–15 wt% HCl or organic acid solution is spotted across the open hole. The reaction yields CO₂ and soluble calcium salts, ideally disintegrating the filtercake structure. The limitation of such treatment is that the acid reacts rapidly and uncontrollably upon contact with the filtercake and formation. This often causes a rapid localized reaction at the entry point, creating a high-permeability channel or pinhole. The acid and completion brine then tend to leak off through that path, leaving much of the filtercake untouched further along the wellbore. The result is a partial and

a non-uniform removal. Moreover, strong acids are highly corrosive to tubing and tools, requiring special metallurgy or corrosion inhibitors. In summary, while acids provide aggressive dissolution, they lack the necessary delay in extended operations and the uniform coverage.

Oxidizers

Peroxides or persulfates target the polymeric component of filtercakes. They break down starch, xanthan, or other organics into smaller fragments. Sometimes oxidizers are included as encapsulated solids in the drilling fluid (to be activated later by an acid trigger) or spotted as a breaker pill (Albonico, et al. 2007). The limitations of such treatments are that traditional oxidizers are highly reactive and non-selective, which means they can attack not only the filtercake polymers but also completion hardware and even formation hydrocarbons. They tend to have a short lifespan at high temperature – thermal decomposition or uncontrolled reaction can consume their activity quickly. Even when "coated" for delay, oxidizers may still react too fast once triggered. Thus, oxidizers are generally unsuitable as a sole solution for high-temperature, long-duration cleanup (Watson, et al. 2010).

Enzymes

Enzyme breakers use biological catalysts (e.g., amylases) to selectively degrade specific polymers in the filtercake, typically starch-based components. Enzymes can efficiently cleave polymer chains into soluble sugars without the harsh effects of acids or oxidizers. Enzyme treatments have been popular for filtercake cleanup in reservoirs where acids are undesirable. Notably, enzymes act slowly and mildly, which promotes a more uniform breakdown of the filtercake. This tends to avoid the sudden creation of fluid loss channels. The limitation of an enzyme treatment is the temperature-sensitive. Most enzymes are limited to low temperatures (Ali A, et al. 2023). Another limitation is that enzymes alone only remove the polymer fraction and do not dissolve carbonate bridging solids. An enzyme-only treatment thus risks leaving behind the solid particles (CaCO_3), which could still impair flow if not addressed.

Chelating Agents

Chelants are chemicals (e.g., EDTA, DTPA, GLDA) that bind metal cations like Ca^{2+} , thereby dissolving calcium-based minerals such as limestone or CaCO_3 bridging material. In filtercake breakers, chelating solutions can gradually dissolve CaCO_3 in the cake without the strong corrosiveness of HCl. They tend to react more slowly than acids, helping with controlled removal. The limitation of chelants is that they are generally less reactive than acids, which is good for delay, but also means they may struggle to dissolve large quantities of CaCO_3 within practical timeframes. High concentrations (10–20 wt%) and extended soaking periods might be needed. Crucially, chelating fluids have no inherent effect on polymers. Furthermore, chelating agents can precipitate or lose effectiveness in high- Ca^{2+} brines or high salinity environments if not carefully formulated (Natalia, et al. 2011). They also exhibit faster kinetics at higher temperatures, so above 200 °F their reaction may no longer be sufficiently retarded.

Acid Precursors

Acid precursors are chemicals that are initially non-acidic or weakly acidic, but slowly generate acid upon hydrolysis in the presence of water and heat. An example is an ester that produces an organic acid like formic or glycolic acid over time. These precursors can be spotted in a completion brine as a neutral solution, which will begin releasing acid in-situ to dissolve CaCO_3 in the filtercake. The delay allows time to deploy the breaker and complete well operations before significant reaction begins. Acid precursors promise uniform filtercake removal by gradually generating acid across the interval. They also tend to be less corrosive initially (K.A. Mahdi, et al. 2014). A key challenge is controlling the hydrolysis rate of the precursor, especially at high temperature. Hydrolysis is temperature-dependent; at temperatures above 200

°F, many low-molecular-weight esters hydrolyze very fast. This can short-circuit the intended delay. By the time placement is done, the acid may already be largely released and spent.

Emulsified Acids in Matrix Stimulation and Potential for Filtercake Cleanup

One approach from matrix stimulation that offers a template for controlled acid reaction is the emulsified acid system. Emulsified acids are essentially acid-in-oil emulsions where the acid is usually 15–28% HCl or an organic acid blend. The acid phase is dispersed as fine droplets within a continuous oil phase using emulsifiers. In matrix acidizing of carbonate reservoirs, such emulsified acids have been used for decades as a retarded acid method. The oil-phase continuous emulsion delays the acid's contact with the formation rock, thereby reducing the acid spending rate and allowing deeper penetration of live acid into the formation before it is consumed. This concept has been successfully applied in both matrix acidizing and acid fracturing to achieve longer etching lengths and more uniform treatment.

Typically, emulsified acid is designed to be 30% oil external phase and 70% acid internal phase by volume. The stability of the emulsion is crucial where it must remain stable during pumping down the well and while reaching the target zone. Once in place, the acid begins to seep out or the emulsion gradually breaks due to elevated temperature, or contact with the formation fluids, slowly releasing acid to react with the carbonate formation. Laboratory studies have shown that emulsified HCl can penetrate further and create deeper wormholes compared to an equal volume of neat HCl.

Given this advantage, emulsified acid system might be repurposed for filtercake removal. The filtercake removal problem shares some similarities with matrix stimulation where both involve delivering acid to react with CaCO_3 in a controlled way, but also important differences. In filtercake cleanup, the acid is mostly reacting with a thin layer of deposited solids of CaCO_3 and polymer on the borehole wall, rather than a deep formation matrix. Filtercake treatment is desired to spend itself within that filtercake layer rather than penetrate beyond it. An emulsified acid could be beneficial here by delaying acid release, ensuring it reacts with the filtercake uniformly along the wellbore length instead of all at one spot. Conventional acids or even standard acid precursors struggle to provide multi-day reaction control at high temperatures, whereas a well-designed emulsion might physically prevent the acid from all reacting at once.

One limitation of standard emulsified acids is the temperature stability over long durations. While they significantly retard acid reaction, emulsions can still break down with time and temperature. Most matrix acid treatments last for hours at most, and not days. Additionally, emulsified HCl alone would only target the carbonate, it would not inherently remove starch or other polymers. Thus, a pure emulsified acid approach to filtercake might result in undissolved polymer residue that still impairs flow.

Novel Approach: Emulsified Acid Combined with Enzyme for High-Temperature Cleanup

To overcome the above challenges, this work explores a hybrid system that combines an emulsified acid with an enzyme in one formulation for filtercake removal. The novelty is applying emulsified acid technology from matrix stimulation to the realm of filtercake cleanup, and incorporating a thermally-stable enzyme to attack the polymer fraction, thereby making the system capable of removing both key filtercake components in a single retarded step. Crucially, the emulsified acid provides a high-temperature delay mechanism that protects the enzyme and controls the acid reaction rate, enabling the combined fluid to work over an extended period of several days even at 200 °F. The acid being used in combination with enzyme is an organic acid which is weaker and slower compared to HCl.

This novel emulsified filtercake removal treatment delays acid spending beyond the traditional 24 hour limit at high temperature, because the organic acid is not free in solution but partitioned in a stable emulsion. This ensures no premature filtercake breakthrough or massive leak-off occurs during the critical well completion period. It also provides uniform coverage; the emulsion contacts the filtercake along the entire open hole and the acid will be slowly released rather than being consumed at the first entry point.

The presence of the enzyme ensures that once the treatment is complete, no polymeric residue remains. The enzyme continuously degrades starch in situ. Moreover, the enzyme is a slow-act additive which can allow for controlled reaction overtime.

The system is designed to be thermally stable. The emulsification system (surfactants, oil phase), the selected enzyme and the organic acid are chosen to withstand 200 °F for days. This extends the applicability of filtercake removal treatments into extended operations without premature fluid loss, which traditionally was problematic.

By keeping the organic acid trapped in an oil phase until it contacts CaCO_3 , the fluid reduces free acid exposure to metal surfaces. Combined with added corrosion inhibitor, this yields very low corrosion rates.

Experimental Procedure and Materials

To investigate the performance of the emulsified organic acid/enzyme filtercake removal system, a series of laboratory experiments were conducted. These experiments covered the preparation of representative WBM filtercake samples, the formulation of the novel emulsified filtercake removal fluid, and the testing of its dissolution efficiency, thermal stability, and corrosivity.

Materials

A water-based mud (WBM), comprising a low-solids, non-dispersed, water-soluble polymer-based drilling mud with acid-soluble bridging and weighting agent was formulated to simulate a reservoir drilling fluid that would be used in an open-hole completion. The bridging agent is calcium carbonate and it was used at a mix of fine and medium-sized CaCO_3 particles to ensure a tight, impermeable filtercake. [Table-1](#) shows recipe of the used drilling fluid in this study.

Table 1—WBM formulation.

Additive Name	Unit	Value	Primary Function
DEFOAMER	GALLON PER BARREL	0.15 to 0.15	DEFOAMER
SODA ASH (Na_2CO_3)	POUNDS PER BARREL	0.25 to 0.5	CONTAMINANT REMOVER
CAUSTIC SODA (NaOH)	POUNDS PER BARREL	0.25 to 0.75	PH ADJUSTMENT
SALT - NaCl	POUNDS PER BARREL	42 to 42	WEIGHTING MATERIAL
XANTHAN GUM MI SCH	POUNDS PER BARREL	1 to 2	VISCOSIFIER
BIO-PAQ AR	POUNDS PER BARREL	6 to 8	FLUID LOSS ADDITIVE
MARBLE FINE TOP PHARM	POUNDS PER BARREL	10 to 10	WEIGHTING MATERIAL
MARBLE MEDIUM MI SCH	POUNDS PER BARREL	5 to 5	WEIGHTING MATERIAL
CaCO_3 25 ARAB. MIN.	POUNDS PER BARREL	10 to 10	BRIDGING MATERIAL
CaCO_3 50 ARAB. MIN.	POUNDS PER BARREL	10 to 10	BRIDGING MATERIAL
RADIAGREEN EME	PERCENT	1 to 3	LUBRICANT
MIL-GARD XPR	POUNDS PER BARREL	2 to 3	H_2S SCAVENGER

Starch polymer was included for fluid loss control. Starch is widely used in WBM and contributes to filtercake structure by plugging pores and acting as a binder. In addition, xanthan gum was used as a viscosifier to achieve appropriate rheology for the drilling fluid. These polymers help form a thin, low-permeability filtercake.

The above drill-in fluid is comparable to those used in field applications for drilling pay zones with minimal damage, containing sized calcium carbonate as a bridging material and starch as a degradable polymer. The fluid was prepared and conditioned prior to making filtercakes.

Filtercake Preparation

Lab filtercakes were generated to simulate the type of damage encountered in the field. API HPHT filter press cells were used to create filtercake on a porous medium.

A 300 mL volume of the prepared WBM drill-in fluid was placed in the HPHT cell. As the filter medium, a porous ceramic disk (1/4 inch thick, 20 microns) was fitted at the bottom of the cell to represent a formation face. This disk simulates a reservoir rock onto which the filtercake would deposit. The cell was heated to 200 °F, and a differential pressure of 300 psi was applied for 2 hours to deposit the filtercake. After 2 hours, the cell was cooled and depressurized, and the disk with the attached filtercake was carefully removed. The resulting filtercake was a few millimeters thick. The filtercake was kept hydrated in a humid chamber to prevent drying out before tests.

Filtercake scrapings were collected to perform weight-based dissolution measurements. This was done by gently scraping the cake into a flask and weighing the filtercake mass. These scrapings contained a representative mix of the cake's CaCO_3 and polymer.

Filtercake Dissolution Tests

The primary experiment to evaluate the emulsified filtercake removal treatment performance was a solubility test over time. Each test involved soaking filtercake material in the emulsified fluid for a specified duration and measuring how much of the solid cake dissolved.

Approximately 10 g of filtercake were placed into a glass bottle inside an aging cell. Then 100 mL of the emulsified acid/enzyme fluid was added, yielding a 1:10 solid:solution ratio by weight. The aging cell was pressurized and sealed to ensure no leakage. The cell was then placed in a convection oven at 200 °F. No rocking or agitation was provided; it was a static soak, which is a conservative scenario. Multiple identical cells were prepared to allow opening them at different time intervals (1 day, 3 days, and 5 days).

The 24-hour point (1 day) serves as a benchmark for conventional systems). The 3 days point indicate how the dissolution progresses in the medium duration, and the 5 days point was chosen as an extreme soak time to test the ultimate capacity and stability of the system.

Thermal Stability Test

A separate experiment addressed the stability of the emulsified filtercake removal fluid itself under prolonged high temperature exposure, without filtercake present. This was to ensure the fluid would not phase-separate prematurely. These stability tests verified that the emulsified filtercake removal fluid system could survive the durations needed for the filtercake dissolution tests.

Corrosion Testing

While prolonged contact of conventional filtercake removal treatment with completion metallurgy is limited in time, it was important to quantify corrosivity of the new system to ensure it would not damage the well tubulars or tools during the soak. Corrosion tests were performed using coupons of C95 steel, a high-strength casing/tubing material often used in completions.

Polished C95 steel coupons were cleaned, weighed, and then placed into the emulsified fluid. Each coupon was completely submerged in ~100 mL of the emulsified acid/enzyme fluid inside a glass bottle. The containers were then sealed and placed in the 200 °F oven for 1, 3 and 5 days.

Results and Discussion

The performance of the emulsified filtercake removal treatment is discussed in terms of filtercake dissolution efficiency, reaction rate control, thermal stability observations, and corrosion impact.

Thermal Stability and Integrity of the Breaker System

The experiments demonstrated that the emulsified acid/enzyme system can withstand high temperature conditions for extended durations without breaking down. No precipitation or incompatibility issues were encountered. The tests showed the emulsion remained intact for 5 days at 200 °F. The ability to hold the acid in an emulsified state for multiple days is a significant finding. Conventional emulsified acids used in stimulation might not be formulated to last that long without phase separation. The use of a robust surfactant package and appropriate phase ratios was key.

The enzyme's survivability at 200 °F, as evidenced by starch degradation tests after aging, was remarkable. This attributes to both the choice of a thermophilic enzyme and the fact that, within the emulsion, the enzyme might be protected by being in a buffered acid phase and possibly not fully active until it contacts the filtercake. Some loss of activity over time is expected, but clearly the enzyme remained active long enough to accomplish its task of polymer removal.

Filtercake Dissolution and Delayed Reaction Performance

After 1 day of soaking, the breaker fluid had dissolved approximately 30% of the filtercake by weight. This indicates the acid reaction was successfully retarded. No sudden loss of whole filtercake chunks or early breakthrough was observed. In conventional acid treatments, almost all acid would have been spent within the 24 hours period, likely creating a localized hole. Here, by 24 h the emulsified acid had only begun to slowly act, showing the intended delay.

After 3 days, dissolution reached about 40%. Significant portions of the CaCO_3 were gone. When the cell was opened at 3 days, it was observed that the filtercake had fragmented into a soft, mushy mixture. The polymer matrix was targeted by the enzyme.

After 5 days, dissolution was ~50%. Only a minor residue of solids was recovered. The treatment fluid by now had turned from emulsion to more of a brine. The continued increase from day 3 to day 5 indicates that fresh acid was still available even in that period which is a sign of prolonged reaction capability.

These results are a dramatic improvement over typical high-temperature filtercake removal treatment performance. Conventional acid treatments rarely maintain activity beyond the first day at 200 °F. Enzymes alone would likely not survive long enough to show additional benefit beyond perhaps 1–2 days. In contrast, this novel emulsified system clearly controlled the reaction rate such that dissolution was spread over 5 days.

Notably, in the combined system, the enzyme likely played a major role in the early stage. First 1–2 days, the enzyme had time to attack the starch in the filtercake as evidenced by the iodine test for the 3 days sample. It is also worth noting that no secondary damage was observed from the emulsified fluid. When the acid spent, the leftover fluid was basically a mixture of diesel, brine, and oil-soluble surfactants. This should be easy to flow out or flush with production. There was no indication of any persistent emulsion or sludge formation. The emulsified fluid, after spending, separated into clear oil and water phases, which is ideal for clean cleanup.

Corrosion and Metals Protection

After the exposure period of 1, 3 and 5 days, the coupons were retrieved, rinsed thoroughly, then dried and re-weighed. The weight loss was used to calculate the corrosion rate. The weight loss of C95 coupons in the emulsified acid/enzyme fluid was very low for the 5-day test and showed no risk to well and completion tubulars.

The encapsulation of acid in oil drastically reduces the immediate acid attack on metals. Even when the fluid is in place across metal (like screens or tubulars) for hours or days, the majority of acid is locked away in droplets. This is a stark contrast to a conventional filtercake removal treatment.

No localized pitting was seen on the coupons. This is important for downhole tools like screens – pitting can weaken them severely even if average corrosion is low. The gentle nature the fluid likely contributed to absence of pitting.

Overall, the corrosion tests reinforce that the emulsified acid/enzyme system is compatible with downhole tubulars. This is a critical consideration where tubular integrity must be preserved.

Comparison to Conventional Systems and Field Implications

A conventional enzyme-only breaker at 200 °F might remove starch but leave CaCO_3 untouched. the emulsified organic acid/enzyme removed both. Moreover, most enzymes could not function long at this temperature. These findings demonstrated that enzyme can be effectively utilized for extended durations at 200 °F when formulated correctly. A conventional acid or acid precursor might have dissolved CaCO_3 quickly but risked an incomplete job due to immediate breakthrough and inability to remove starch.

Conclusions

This study introduced and validated a novel filtercake breaker fluid that combines an emulsified organic acid system with an enzyme for use in water-based mud drilling operations, particularly targeting high-temperature open-hole completions. Based on the extensive literature review and laboratory evaluation, the following conclusions were drawn:

1. Conventional filtercake removal methods have critical limitations in high-temperature environments. Strong acids react too fast and non-uniformly, risking partial cleanup and high corrosion. Enzymes alone cannot withstand high temperatures and do not remove mineral components. Chelating agents are slow and can be incompatible with polymers or ineffective at breaking starch. Acid precursors provide some delay but typically fail to maintain controlled release beyond ~24 hours at 200 °F, leading to premature spending. These limitations necessitate either multi-stage treatments or accepting suboptimal cleanup in high temperature wells.
2. Emulsified acids, widely used in matrix stimulation to retard acid reaction, can be successfully adapted for filtercake removal when coupled with an enzyme. The emulsified acid acts as a delivery vehicle for acid that significantly delays its reaction with the filtercake, enabling prolonged, uniform acid spending. This is a novel application of emulsified acid beyond its traditional use for wormholing in carbonate formations.
3. The integrated enzyme in the breaker fluid addressed the polymer component of the filtercake, resulting in comprehensive cleanup. The polymer breakdown enhanced overall removal by destroying the filtercake's structural integrity, allowing acid to contact and dissolve carbonate particles more thoroughly. The combination achieved what neither acid nor enzyme could do alone in a one-step removal of both starch and CaCO_3 .
4. Laboratory solubility tests demonstrated extended reaction control and high filtercake dissolution efficiency of the new within a 5-day period at 200 °F. Such multi-day, controlled dissolution at 200 °F far exceeds the performance of conventional breaker systems, which often complete reaction in less than 24 hours.
5. The novel breaker fluid exhibited excellent thermal stability at 200 °F. The acid-in-oil emulsion remained intact for several days under static conditions, and the enzyme retained activity over the required timeframe. There were no signs of emulsion collapse or phase separation until after the majority of the filtercake had been removed. This stability is critical for ensuring the breaker's longevity in downhole applications and was achieved through a robust surfactant package and phase ratio selection.
6. Corrosion tests on C95 steel showed minimal metal loss, confirming the system's compatibility with downhole tubulars and tools. The encapsulation of acid in the oil phase minimized the corrosivity risk of the fluid. Measured corrosion rates were extremely below what would be expected from a direct acid treatment. This indicates that the emulsified acid/enzyme breaker can be safely deployed in completion operations without significant risk to completion hardware.

7. The emulsified acid/enzyme system enables uniform filtercake removal along the wellbore. By avoiding instantaneous acid leak-off, the breaker fluid had time to contact the entire filtercake area. The gradual, evenly-distributed reaction will lead to a homogeneous cleanup. This uniformity is expected to translate to improved well productivity or injectivity.

In conclusion, the emulsified organic acid + enzyme filtercake removal system represents a significant advancement in wellbore cleanup technology. Field implementation of this system can be done with conventional pumping equipment and straightforward pill placement. Future work may include field trials of this breaker fluid, as well as investigations into optimizing the surfactant or enzyme selections for even higher temperatures or different filtercake compositions. The encouraging findings here strongly support moving toward such field applications, where the novel breaker can solve longstanding issues of delayed, uniform filtercake removal. One challenge that was carefully addressed in is ensuring the emulsion stability in presence of the filtercake. If the emulsion breaks too early, it could release acid prematurely. The lab tests involved contact with the filtercake and the fact that delayed reaction was observed suggests that the surfactant did its job and the emulsion didn't break upon touching filtercake. Overall, the experimental results strongly support the feasibility of using an emulsified acid + enzyme system for extended duration high-temperature filtercake removal operations.

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